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2027 CYCLE

MOCK · PRACTICE · CRACK

CUET UG

General Aptitude Test

Scientific Literacy & General Mental Ability

Official Syllabus · GT Paper · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA official CUET(UG) 2026 syllabus — General Aptitude Test (501), the current 2027 baseline)	Skills the questions test
Everyday science	Everyday application of science	Explaining the science hiding inside daily life — kitchens, weather, health.
Observation	Observation	Noticing what is given, what changed, and what stayed fixed in a scenario.
Judgement	Judgement	Evaluating claims on evidence and mechanism, not on confidence or drama.
Problem solving	Problem solving	Running a clear loop: define, try the simplest path, verify, conclude.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each test paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The General Aptitude Test section named in the official syllabus is **Scientific Literacy and General Mental Ability**: everyday application of science, observation, judgement and problem solving. No degree-level physics or chemistry is needed — the questions ask whether you can explain the world you already live in, with mechanisms in plain words.

What Scientific Literacy Actually Tests

Read the syllabus line again: **everyday** application of science. This section does not want thermodynamics — it wants to know whether you can say WHY a pressure cooker cooks faster, WHY ice floats on water, and whether you can smell a fake health claim before it reaches your wallet. The tested skills are thinking habits: observe carefully, explain with mechanisms, demand evidence, and solve step by step. A student who understands ten everyday mechanisms deeply beats a student who half-remembers a hundred textbook paragraphs.

- **Mechanisms, not verdicts.** "Ice floats" is a fact; "ice floats BECAUSE frozen water is less dense" is literacy. Every answer in this section lives in the BECAUSE.
- **Plain words are allowed** — encouraged, actually. If you cannot explain a mechanism to a Class 6 student, you have not understood it yet.
- **New scenarios, old mechanisms.** The question may describe a situation you have never seen — but the mechanism behind it will be one you have. Match the mechanism, not the surface story.
- **Judgement questions test your defences:** against miracle cures, superstitions dressed as science, and conclusions jumped from coincidence.
- **The ladder is your universal tool:** observe → question → explain → test → conclude. It runs every question in this section.

REAL-WORLD ANALOGY

Scientific literacy works like a **mechanic's ear**. A mechanic hears an engine and does not just say "it's broken" — he locates the cause: "the belt slips when cold, so it squeals until it warms." Same engine, same sound — but the cause carries the fix. Every everyday-science question is that squeal: a familiar phenomenon asking for its cause in plain words.

Everyday Science Around You: The Anchor Set

These ten mechanisms cover a remarkable share of everyday-science questions. Learn each one as a STORY — phenomenon, then cause in one breath — and practise saying the cause aloud without the phenomenon in front of you. That is the whole game.

Everyday sight	The mechanism, in plain words	The trap nearby
Lightning before thunder	Light travels far faster than sound, so the flash arrives first	"Sound is louder" — no; it is about speed
Ice floats on water	Water expands on freezing, so ice is LESS dense and sits on top	"Ice is heavier" — exactly backwards
Pressure cooker cooks faster	Higher pressure raises water's boiling point above 100°C, so food meets hotter water	"Seals the heat in" — vague; the boiling point is the cause
Sweating cools the skin	Evaporating sweat carries heat away from the body	"Sweat is cold" — it BECOMES cooling by evaporating
Salt on icy roads	Salt lowers the freezing point, so ice melts even below 0°C	"Salt warms the road" — no; it changes freezing point
Metal spoon feels hot in soup	Heat travels along the metal by conduction	"The soup soaks in" — no flow of liquid involved

Everyday sight	The mechanism, in plain words	The trap nearby
Green leaves	Chlorophyll absorbs red and blue light and REFLECTS green	"Leaves make green light" — they bounce it
Candle dies under an inverted glass	The flame consumes the trapped oxygen and then starves	"The glass sucks the flame out" — nothing sucks
Hot air rises	Warmer air expands, becomes less dense, and floats above cooler air	"Heat pushes up" — density does the lifting
Vaccination protects	A weakened or harmless form trains the immune system BEFORE real infection	"Vaccine gives the disease" — the training form is tamed

Observation: Reading the Scene Like a Scientist

Observation questions hand you a scenario — a described experiment, a photo setup, a small story with data — and ask what you can conclude. The failure mode is answering from memory instead of from THE SCENE. The method: treat the question like a crime scene, and inventory it before theorising.

- **Step 1 — List the givens.** What is present? What numbers, objects or conditions are stated? Write them as a bullet list.
- **Step 2 — Mark what changed and what stayed fixed.** Change is where the story lives; the fixed items are the controls.
- **Step 3 — Ask the one question the item asks.** Restate it in your own words: "WHY did the water level change?" beats a vague stare at four options.
- **Step 4 — Sketch if motion or arrangement is involved.** A ten-second diagram of the setup exposes relationships that prose hides.
- **Step 5 — Conclude only from the list.** If an option needs facts NOT in the scene, it is out of scope — exactly like RC elimination.

SOLVED OBSERVATION — THE FLOATING ICE CUBE

Scene: A glass is filled to the brim with water. Ice cubes float in it, partly above the surface. The ice melts fully. Question: does water overflow the brim, fall below it, or stay level?

- **Givens:** ice floats (so ice is less dense); it displaces water equal to its own WEIGHT while floating.
- **Mechanism:** the space the ice occupies UNDER the surface holds exactly the water the melted ice will add — floating ice displaces its weight in water, and melted ice becomes exactly that weight of water.
- **Conclusion:** the level STAYS at the brim — no overflow, no drop.
- **The trap avoided:** "melting adds ice-water, so it overflows" ignores the under-surface space the ice was already occupying. The scene's own facts decide.

Judgement: Evaluating Claims Without Being Fooled

Judgement questions present a claim — a health tip, a "scientific fact", a news-style conclusion — and ask which response is most reasonable. The tested skill is **structured scepticism**: not cynicism, not blind belief, but a fixed checklist run against every claim. Advertising and rumour survive on the hope that you will skip the checklist.

Checklist item	The question to ask	Alarm signs
Evidence	Is real data offered — measurements, trials, records — or only promises?	"Everyone knows", a single anecdote as "proof"
Mechanism	Can the claim say HOW it works, step by step, in plain words?	Vague energy talk; "detox" with no named toxin
Source and motive	Who gains if you believe — expertise or profit?	The seller is also the evidence's only source
Alternatives	Could something simpler explain the same result?	A dramatic cause claimed where a boring one suffices
Framing	Are the words measured or theatrical?	"Instant", "cures everything", "no side effects", "secret"

- **Correlation is not causation — the classic judgement trap.** Ice-cream sales and drowning deaths rise together every summer; ice cream does not cause drowning — summer causes both. Whenever two things rise together, hunt for the hidden third factor.
- **The reasonable-response pattern:** when options ask "what should one conclude?", the answer backed by evidence + mechanism + no better alternative wins — and "more research is needed before believing" is a perfectly respectable answer to an unevidenced claim.

SOLVED JUDGEMENT — THE FEVER PILL

Claim: "This herbal pill cures any fever in a single dose, with no side effects."

- **Evidence?** None offered — only the promise. **Mechanism?** Missing; fevers have many different causes, so one dose curing ALL of them has no plausible how.
- **Framing?** "Any" + "single dose" + "no side effects" — three alarm words in one sentence.
- **Most reasonable response:** reject the claim UNTIL controlled evidence with a stated mechanism appears.
- **The trap:** "herbal means safe" — source does not replace evidence; the checklist, not the label, decides.

Problem Solving: The Four-Step Loop

Problem-solving questions are mini-puzzles: a goal, some constraints, and options that all look plausible. The winning habit is a loop, run on paper: **define the actual goal in one sentence → list the simplest approaches → try the cheapest one fully → verify against every condition**. The loop prevents the two classic failures: solving a DIFFERENT problem than the one asked, and stopping at the first answer that merely looks nice.

- **Step 1 — Define.** "What exactly must be true at the end?" Say it as one sentence. Half of all wrong answers solve a slightly different question.
- **Step 2 — Generate approaches briefly.** Two or three candidate paths, not twenty. Prefer the one using the fewest assumptions.
- **Step 3 — Execute the simplest path fully.** No skipped lines — the same discipline as worked maths examples.
- **Step 4 — Verify against every condition.** Tick each constraint; if one fails, the approach is wrong, not the puzzle.

SOLVED PROBLEM SOLVING — THE FIVE-LITRE POUR

Puzzle: You have a 5-litre jug, a 3-litre jug, and an endless tap. Measure exactly 4 litres.

- **Define:** end state = exactly 4 L sitting in the 5 L jug.
- **Approach 1 (fill-and-transfer):** fill the 5 L jug → pour into 3 L jug, leaving 2 L in the big jug → empty the 3 L jug → pour the 2 L into the 3 L jug → fill the 5 L jug again → top up the 3 L jug (takes 1 L, since 2 L are already there) → the big jug now holds $5 - 1 = 4$ L.
- **Verify:** every pour is legal; the big jug holds exactly 4 L. All conditions ticked.
- **The trap:** trying to see the answer by imagination instead of simulating pours — the loop on paper beats the guess in the head, every time.

Key Rules and Strategies: One-Stop Revision

SCIENTIFIC LITERACY & GMA — EVERY TOOL ON ONE PAGE

Tool	Statement	Exam use
The ladder	Observe → question → explain → test → conclude	Every scenario question
Mechanism rule	An answer must say HOW, in plain words	Everyday-science questions
Claim checklist	Evidence, mechanism, source, alternatives, framing	Judgement questions

Tool	Statement	Exam use
Alarm words	"Instant", "cures all", "no side effects", "secret" — suspicion triggers	Fake-claim questions
Correlation $\neq$ causation	Two things rising together needs a mechanism or a hidden third cause	Conclusion questions
Out-of-scope rule	An option needing facts absent from the scene is out	Observation questions
Fixed vs changed	Inventory givens; the change carries the story	Scenario reading
Sketch habit	Draw setups and flows in ten seconds	Motion and arrangement puzzles
Simplest path	Execute the cheapest approach fully before fancy ones	Problem-solving options
Verify every condition	Tick all constraints; one miss = wrong approach	Puzzle questions
Health anchors	Vaccines train immunity; sunlight gives vitamin D; standing water breeds mosquitoes	Health and environment items
Marking maths	+5 right, −1 wrong — mechanism-backed attempts only	Attempt discipline

Common Errors and Special Cases

THE TRAPS THIS TOPIC SETS — READ BEFORE THE MOCK

The habit you assume	The twist	The trap in the question
"Memorised verdicts are enough"	The question wants the MECHANISM, the because	Everyday-science options
"Heat and temperature are the same"	Heat is energy in transit; temperature is a reading	Kitchen-science questions
"Heavier objects fall faster"	Without air resistance, mass drops at the same rate	Falling-body scenarios

The habit you assume	The twist	The trap in the question
"Ice is heavier than water"	Water expands on freezing — ice is LESS dense, so it floats	Floating and level puzzles
"Confident claims are true claims"	Confidence is free; evidence and mechanism are not	Miracle-cure questions
"Two rising trends imply cause"	A hidden third factor often drives both	Correlation traps
"Herbal or natural means safe"	Source changes nothing — the checklist decides	Health-claim questions
"Answers may use outside facts"	The scene is the only evidence — out of scope otherwise	Observation questions

The 60-Second Self-Test

Cover the answers, say each one aloud, then check. If any answer takes longer than ten seconds, revisit that section.

- **Why does lightning arrive before thunder?** → Light travels far faster than sound.
- **Why does a pressure cooker cook faster?** → Higher pressure raises water's boiling point above 100°C.
- **Name the five rungs of the ladder.** → Observe, question, explain, test, conclude.
- **Name the five claim-checklist items.** → Evidence, mechanism, source, alternatives, framing.
- **Ice-cream sales and drownings rise together — what do you hunt for?** → The hidden third factor (summer) — correlation is not causation.
- **A melting ice cube in a brim-full glass — overflow, drop, or level?** → Level — floating ice displaces exactly its own weight.

The Whole Chapter in One Picture

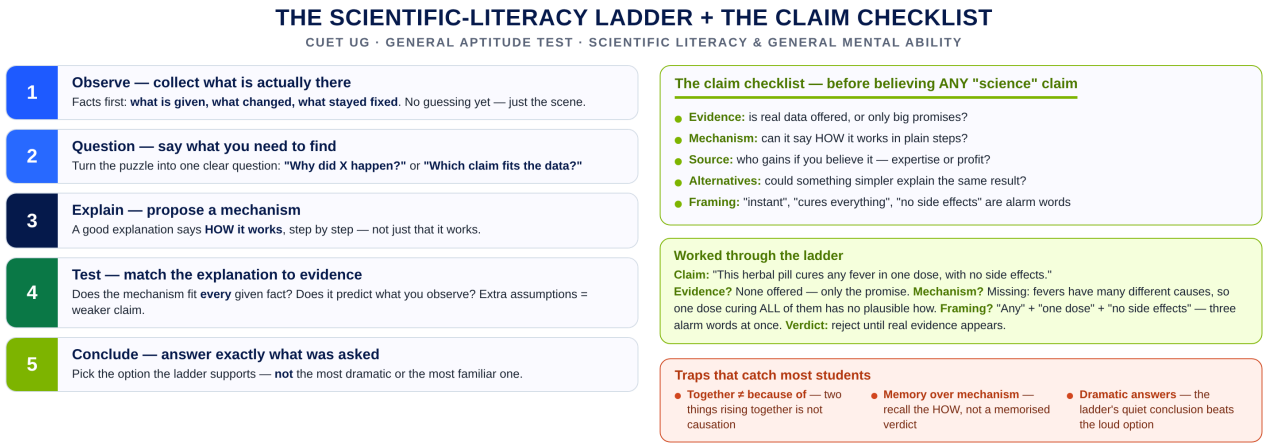


Figure 1: The scientific-literacy ladder — observe, question, explain, test, conclude — beside the claim checklist, a worked claim evaluation, and the traps that catch most students.



Figure 2: Chapter mindmap: everyday-science anchors, the observation method, the claim checklist, the problem-solving loop, correlation-vs-causation, and the health basics that anchor judgement questions.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style — everyday mechanisms, one observation puzzle, and two judgement calls. Every answer must carry its mechanism. Each correct answer earns +5; each incorrect one costs −1.

Q1. We see lightning before we hear thunder because:

(A) thunder takes time to grow loud
(B) light travels much faster than sound
(C) clouds absorb sound waves
(D) lightning happens first in time

Q2. Ice floats on water because:

(A) ice is heavier than water
(B) ice is less dense than water
(C) water pushes ice up with heat
(D) ice contains trapped air only

Q3. Food cooks faster in a pressure cooker because:

(A) the sealed lid keeps all flavours inside
(B) higher pressure raises the boiling point of water above 100°C
(C) the cooker heats food from all sides
(D) steam is hotter than any flame

Q4. Sweating cools the body because:

(A) sweat is colder than blood
(B) evaporation of sweat carries heat away from the skin
(C) sweat blocks sunlight
(D) sweat reflects heat like a mirror

Q5. A message claims: "One pill instantly cures all fevers with no side effects." The MOST reasonable response is to:

(A) forward it — herbal means safe
(B) reject it until real evidence and a mechanism are shown
(C) try one dose to test it personally
(D) believe it, since it sounds scientific

Q6. A metal spoon left in hot soup feels hot at the handle because heat travels by:

(A) conduction along the metal
(B) convection in the spoon
(C) radiation from the bowl
(D) evaporation of the soup

Q7. Leaves look green because chlorophyll:

(A) produces green light
(B) absorbs green light strongly
(C) reflects green light while absorbing red and blue
(D) stores sunlight as green pigment

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Q8. A burning candle covered by an inverted glass goes out because the flame:

(A) is crushed by the glass
(B) runs out of oxygen inside the glass
(C) melts the wax too fast
(D) gets too cold under glass

Q9. Spreading salt on icy roads melts the ice because salt:

(A) heats the road surface
(B) lowers the freezing point of water
(C) absorbs the ice chemically
(D) blocks cold air from below

Q10. A city records that mosquito-disease cases rise every year in the weeks after heavy rains. The BEST-supported conclusion is that:

(A) rain directly causes the diseases
(B) standing water after rains provides mosquito breeding sites, raising cases
(C) mosquitoes become stronger in rainy weeks
(D) hospitals report more carefully after rains

Detailed Solutions

Each solution shows the working, not just the verdict — cover the answer line, attempt the reasoning, then compare.

Sol. Q1. The flash and the crack are born together; light outruns sound by a huge margin, so the flash wins the race to your eyes. "Happens first in time" is the tempting nonsense option.

Answer: (B)

Sol. Q2. Water expands as it freezes, so ice is LESS dense than liquid water and floats. Option A is the exact reverse; trapped air is not the cause.

Answer: (B)

Sol. Q3. Pressure raises water's boiling point above 100°C, so the food sits in hotter water and cooks faster. "Keeps flavours in" may be true of lids generally — but it is not the cooking-speed mechanism.

Answer: (B)

Sol. Q4. Turning liquid sweat into vapour takes energy, and that energy is drawn as heat from your skin — evaporation is the cooling engine.

Answer: (B)

Sol. Q5. Run the checklist: no evidence, no mechanism, three alarm words ("instantly", "all fevers", "no side effects"). The reasonable response demands evidence first. Testing it personally on a real fever is exactly what the claim wants you to do.

Answer: (B)

Sol. Q6. Metal is a conductor: heat flows atom to atom along the spoon from the hot end to the handle. The soup stays put — nothing convects inside solid metal.

Answer: (A)

Sol. Q7. Chlorophyll absorbs red and blue wavelengths for photosynthesis and bounces green back to your eyes — we see the reflected colour, not an emitted one.

Answer: (C)

Sol. Q8. The glass traps a fixed pocket of air; the flame consumes its oxygen, and once oxygen runs short, combustion stops. Nothing "crushes" or "freezes" the flame.

Answer: (B)

Sol. Q9. Salt dissolves into the surface water and lowers its freezing point, so ice melts even when the road stays below 0°C — no heating involved.

Answer: (B)

Sol. Q10. Mechanism-backed conclusion: rains leave standing water — the classic breeding site — so mosquito numbers, and the diseases they carry, rise in the following weeks. Option A names a cause with no mechanism; option D invents a reporting story absent from the data.

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option D	Confusing the race with the schedule	Flash and crack are simultaneous — light simply travels faster
Q2 option A	Trusting the "icy = heavy" intuition	Water EXPANDS on freezing — ice is less dense, so it floats
Q3 option A	Settling for a true-but-vague statement	The mechanism asked is the boiling-point rise, not general lid virtues
Q5 option A	Letting "herbal" bypass the checklist	Source changes nothing — evidence and mechanism decide
Q7 option B	Inverting reflection and absorption	Chlorophyll REFLECTS green; it absorbs red and blue
Q10 option A	Reading correlation as direct cause	Trace the mechanism: standing water → breeding → cases

Your Daily Curiosity Drill

Scientific literacy compounds the way GK does — small daily exposures, noted down. The drill takes ten minutes and needs nothing but the world outside your window. It builds the exact muscle this section tests: seeing a phenomenon and producing its mechanism in plain words.

Minute	The drill	The standard to reach
1-3	Pick ONE everyday phenomenon you met today (steam, a shadow, a rusting nail) and say its mechanism aloud	A because-sentence with HOW inside it
4-6	Test one claim you meet online against the five-item checklist	Verdict: evidence, mechanism, source, alternatives, framing — named one by one
7-8	Spot one correlation in the news and hunt the hidden third factor	A named third cause, or an honest "none visible yet"
9-10	Redraw one mechanism from this PDF from memory, then check against the book	Ten mechanism cards owned, not read

- **Keep a mechanism notebook:** one line per phenomenon — the sight, then the because. Twenty lines a week is a complete revision deck.
- **Re-read the notebook, not textbooks, the night before the exam** — your own collected mechanisms are the highest-value page you own.

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